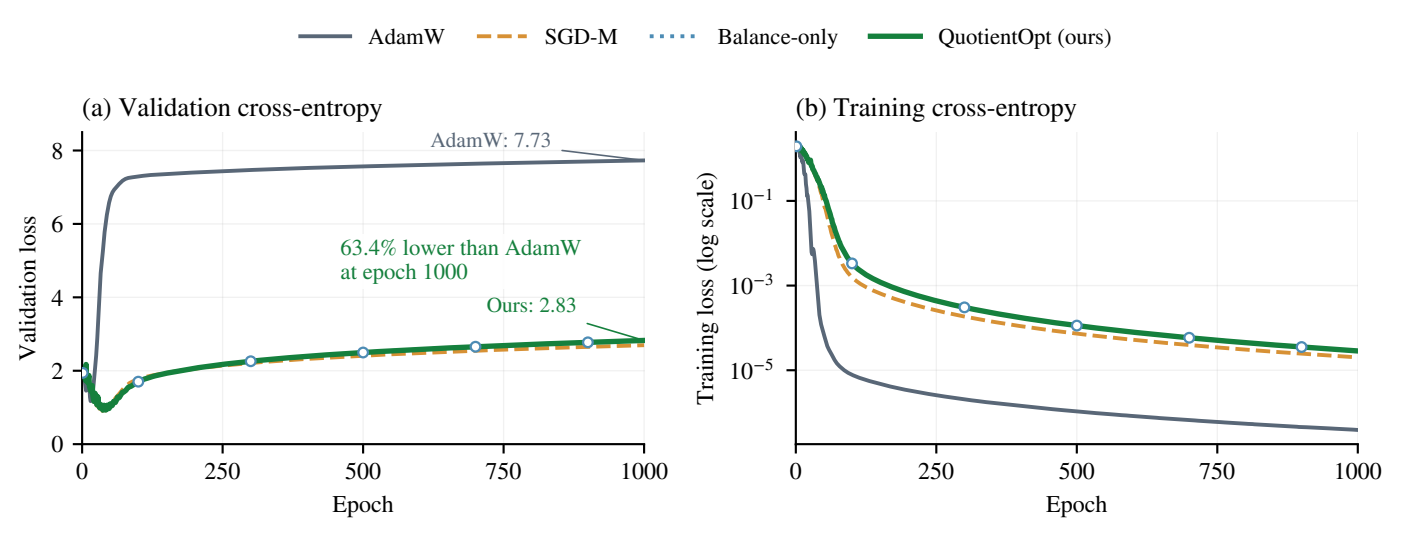
AdamW
SGD-M
Balance-only
QuotientOpt (ours)
(a) Validation cross-entropy
AdamW: 7.73
63.4% lower than AdamW
at epoch 1000
Ours: 2.83
Validation loss
Epoch
(b) Training cross-entropy
Training loss (log scale)
Epoch